

Ayush Bhartal

B.Tech Computer Science & Engineering (2024-2028) | FY CGPA: 9.02 | Sem III SGPA: 8.90

Third year B.Tech Computer Engineering student passionate about AI/ML and software development, with experience building projects involving digital twins, audio processing, and data driven systems.

+ 91 9130196188

° bhartalayush@gmail.com

° Pune, India – 411030

°LinkedIn: [ayush-bhartal-8918103a5](#)

° GitHub: [bhartalayush](#)

EDUCATION

B.Tech in Computer Science & Engineering (Pursuing)

Deccan Education Society, Pune University

Expected Graduation: 2028 | CGPA: 9.02

TECHNICAL SKILLS

- **Programming:** C, C++, Java, Python, C#(Unity)
- **Data & ML:** Pandas, NumPy, Matplotlib
- **Databases:** MySQL
- **Web / UI:** HTML, Java Swing, CSS
- **Core CS:** OOP, DBMS, COA, Discrete Mathematics
- **Tools:** Git, Tableau, LaserCAD, Google AI Studio, n8n, Ultimaker Cura, Unity Engine

PROJECTS

Speech Analysis & Communication Assessment System | Python, Pandas, Faster Whisper

- Developed a speech analytics pipeline for transcription, pause detection, and fluency assessment.
- Extracts communication metrics from recorded audio using local AI models.

Terracotta Pot Digital Twin & Predictive Cooling System | Unity, C#, Python

- Developed a digital twin using real environmental datasets to simulate and predict water cooling behavior.
- Integrated data driven visualization and predictive temperature modeling.

Flood Simulation and Risk Visualization System

- Simulated flood spread using terrain and rainfall parameters to visualize risk zones and impact severity.
- Focused on data driven modeling and visual interpretation for disaster awareness.

Real Time Chord Detection and Harmonic Analysis Tool

- Building a system to detect musical chords from audio input and classify harmonic states in real time.
- Maps detected chords to dynamic lighting patterns, linking harmonic mood to visual output.

PRAVAH – Smart Traffic Management System (SIH 2025):

- Conceptualized an AI-enabled, weather-resilient adaptive traffic system for Odisha with emergency vehicle prioritization and festival-aware signal control.
- Tech Exposure: Python, basic ML, computer vision, edge computing concepts.

Plagiarism Case Management System — Database Project (Team of 4)

- Designing and normalized a relational database for tracking plagiarism cases and review workflows.
- Tech: MySQL or PostgreSQL, HTML, CSS, JavaScript.

LEADERSHIP & ACTIVITIES

Music Head — University Team, Firodia Karandak 2025

- Led and coordinated the university music team for the prestigious Firodia Karandak competition.

Cyber Security Awareness Initiative — NGO

- Conducted awareness sessions for underprivileged teenagers on online scams and cyber fraud prevention.

Technical Team Member | AWS Cloud Club

- Contribute to technical workshops, cloud computing initiatives, and collaborative development projects

LANGUAGES

English (Fluent), Hindi (Fluent), Marathi (Fluent), Telugu (Fluent), German (A2 in progress)

INTERESTS

Algorithmic Problem Solving, Data Analysis, Machine Learning, Computer Vision, AR/VR Simulations (Unity)